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Investigation of the development of the Robotic Dexterous Hands

A Deep Dive into Mechanics, Transmission, and Actuation

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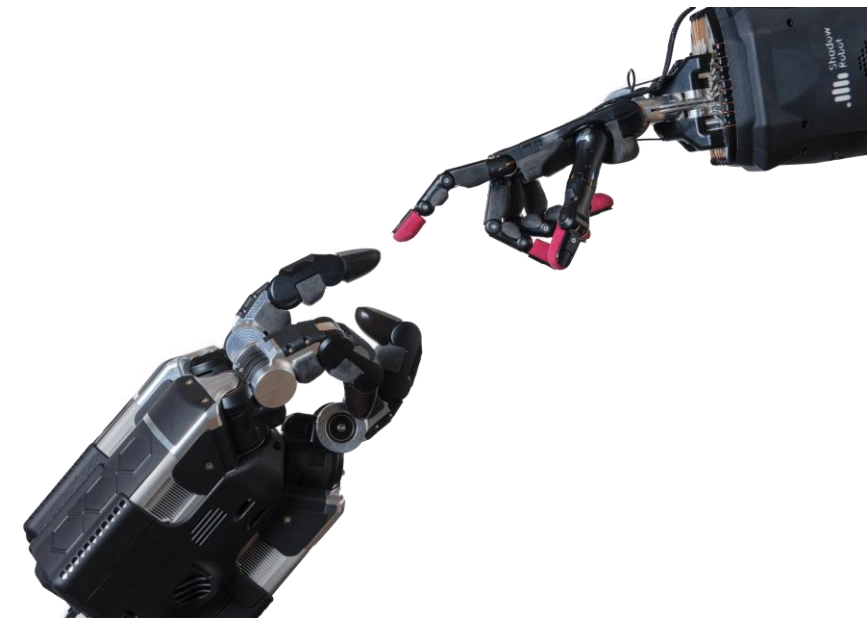


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The intelligence of the dexterous hand is manifested in the synergy between its hardware and software.

- Basic Mechanical design
- Actuation and Transmission
- Sensing Capacity
- Materials and manufacturing
- Control and High-Level AI



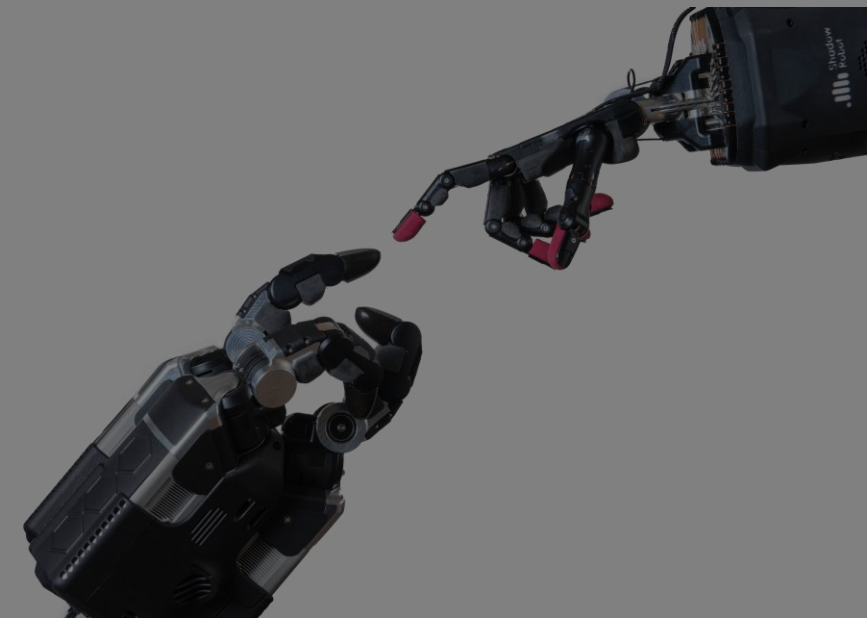


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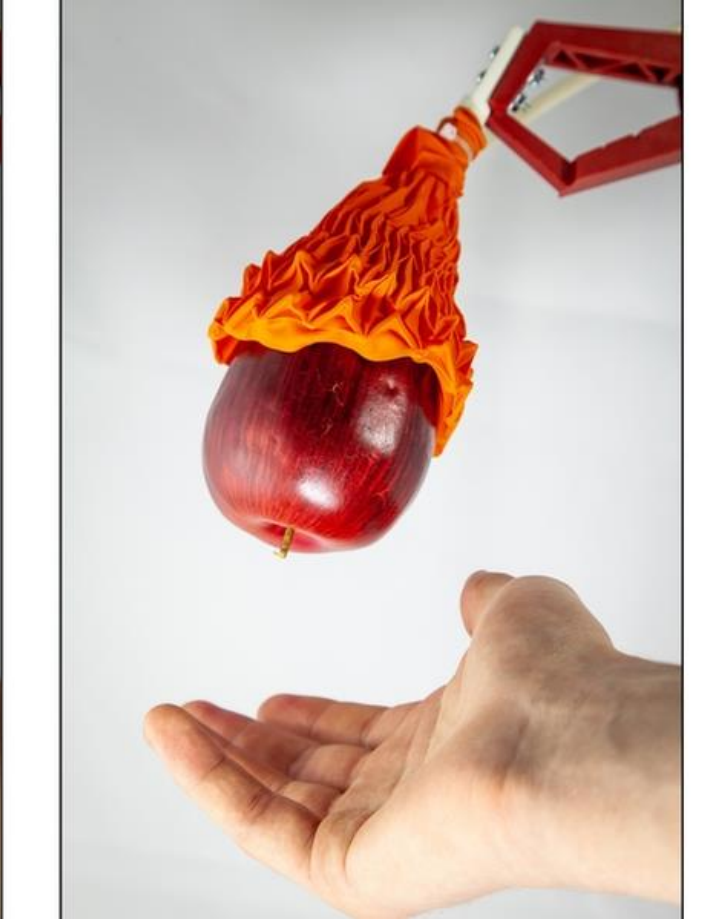
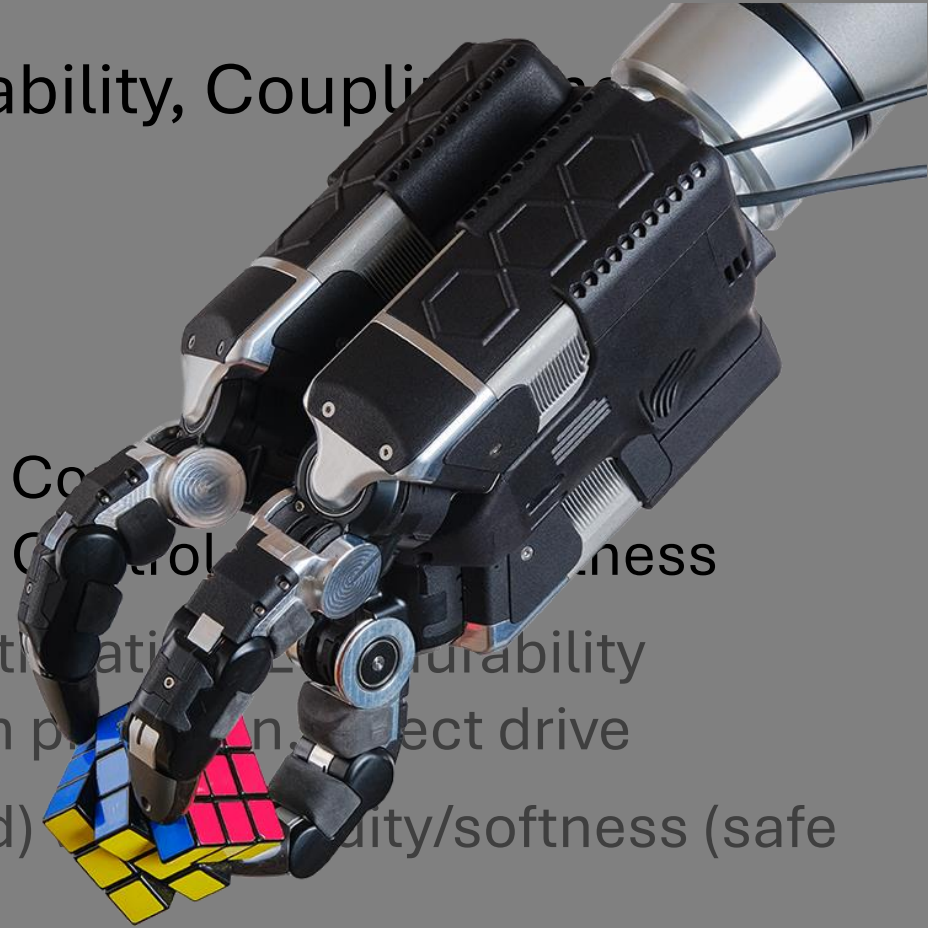




Mechanical design: The Trade-off Between Adaptability, Coupling and Rigidity

The mechanical structure presets the hand's movement patterns and ability boundaries

- Adaptability: Fully actuated - High precision, High cost, Control complexity
Underactuated - Passive adaptive, Simple Control, High Robustness
- Coupling: Tendo-drive – High flexibility, High space utilization, Low durability
Linkage-driven – Mature, High rigidity, High precision, Direct drive
- Rigidity Balance: High rigidity (high precision, large load) vs. low rigidity/softness (safe interaction)

A close-up photograph of a robotic hand, likely a Shadow Hand, holding a Rubik's cube. The hand is black and silver, with visible joints and sensors. The Rubik's cube is partially solved, showing blue, yellow, and red faces. The background is a solid grey color.



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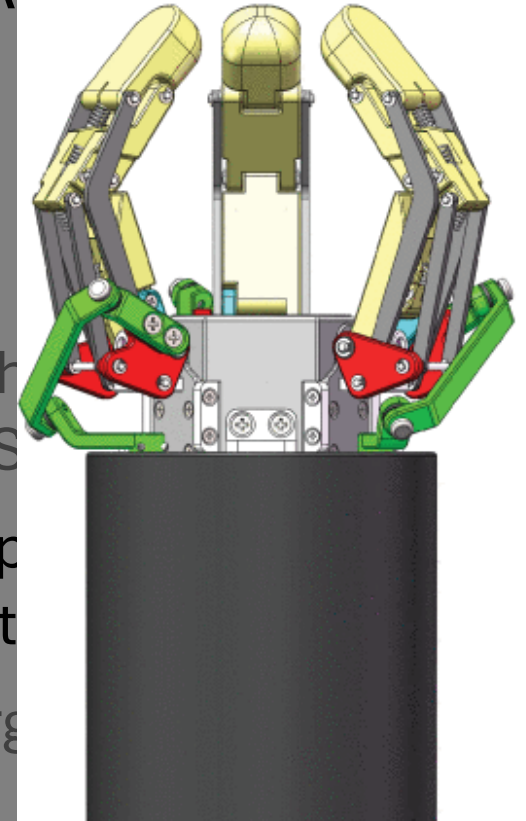
The mechanical movement pattern



the hand's boundaries

- Adaptability:
- Coupling:
- Rigidity Balance (interaction)

High precision, High
passive adaptive, S
flexibility, High sp
ature, High rigidity
High precision, large



Complexity
Robustness
durability
direct drive
ity/softness (safe



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Transmission system - transmitter of power and motion

Transmission is the bridge that converts motor movement into joint action

Tendo – drive

Advantages: High flexibility, High space utilization

Disadvantages: Lower lifespan and Reliability

Representative: Tesla Optimus Hand

Linkage/Gear drive

Advantages: High load capacity, High precision, Mature and commonly used

Disadvantages: Not suitable for High-speed movement

Representative: Inspire Robots (因时机器人)



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Drive Motor – The “muscles” of the dexterous hand

The motor is the source of power, determining performance and size

Rotary motor + Reducer: Most Common scheme, through gears to reduce speed and increase torque

Linear screw servo: High efficiency, high integration

Representative: The core miniature linear servo actuators of Inspire Robots (因时机器人)

Worm gear motor: Compact structure, high reduction ratio, self-locking

Representative: Tesla Optimus 1st generation hand transmission solution



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Marketing

In 2024, the global multi-finger dexterous robot market is valued at approximately **\$2.02 billion**. The expected compound annual growth rate (CAGR) is projected to be above 26% to 39.2%, with the market size reaching around **\$15.26 billion** by 2030

The Chinese humanoid robot industry is expected to expand from approximately **5.3 billion yuan** in 2025 to 38.7 billion yuan in 2028



Summary: The Mechanical Foundation Defines the Path

The synergy of Mechanics, Transmission, and Actuation forms the indispensable physical basis for dexterous hand intelligence

Mechanical

- Adaptability
- Coupling
- Rigidity

Transmission

- Tendo-Drive
- Linkage-Driven

Actuation

- Rotary Motors
- Linear Servos
- Worm Gear

Market validation

Driven by the booming humanoid robot market (CAGR > 26%), optimizing this hardware stack has never been more critical

Future Challenges

The key challenge lies in balancing performance, cost, and reliability, which highlights the need for continuous innovation in modularization and robust design



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